

DA5402W MLOPS ASSIGNMENT

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Developing an end-to-end MLOps pipeline gave me a realistic look at what it takes to move a machine learning model out of a local workspace and into a containerized, production-style setup. Out of everything in the project, the hardest part by far was dealing with the infrastructure side of things, specifically getting Docker and Kubernetes to work together.

Writing the PyTorch model code and setting up the FastAPI inference script wasn't too bad, since that's mostly standard development work. The real bottleneck started when I had to write out all the multi-stage Dockerfiles and YAML configuration files for Kubernetes. Trying to troubleshoot why a pod was failing to start up or tracking down a minor syntax error in a config file meant constantly checking container logs and iterating on the manifests.

On top of the basic deployment, getting the resource configurations right was pretty tricky. Setting up the training job to request GPU resources properly while also configuring the Horizontal Pod Autoscaler (HPA) for the serving application required a lot of careful tuning. Balancing those limits without breaking the service or overwhelming the cluster took multiple rounds of rebuilding images and reapplying manifests.

While the troubleshooting process was tedious at times. Seeing the pods shift to a running state, watching the autoscaler respond, and successfully testing live predictions through the port-forwarded service made all the trial and error worth it.

```
PS C:\Users\HP\mlops-pytorch-pipeline> kubectl get all -n ml-training
NAME                                READY   STATUS    RESTARTS   AGE
pod/ml-training-job-b2fdn           0/1     Pending   0           17m
pod/model-serving-76c48db7df-jf25c 1/1     Running   1 (16m ago) 17m
pod/model-serving-76c48db7df-zh96g 1/1     Running   1 (16m ago) 17m

NAME                                TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
service/model-serving               ClusterIP     10.96.202.2   <none>         80/TCP      17m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/model-serving        2/2     2             2           17m

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/model-serving-76c48db7df 2         2         2       17m

NAME                                REFERENCE          TARGETS          MINPODS   MAXPODS   REPLICAS   AGE
horizontalpodautoscaler.autoscaling/serving-hpa  Deployment/model-serving  cpu: <unknown>/80%  2         5         2           16m

NAME                                STATUS    COMPLECTIONS   DURATION   AGE
job.batch/ml-training-job            Running   0/1             17m        17m

PS C:\Users\HP\mlops-pytorch-pipeline>
```

```
> kubectl port-forward svc/model-serving 8080
Forwarding from 127.0.0.1:8080 to 10.96.202.2:80
Use Ctrl+C to stop the port forwarding.
```

```
> curl -X POST http://localhost:8080/predict -H 'Content-Type: application/json' -d '{"classroom": "test_3stage_pg", "probability": 0.355, "automobile": 0.5518, "size": 0.075, "lat": 0.0852, "den": 0.008, "deg": 0.0011, "freq": 0.0036, "hola": 0.0003, "viva": 0.0066, "track": 0.0727}'
{"classroom": "test_3stage_pg", "probability": 0.355, "automobile": 0.5518, "size": 0.075, "lat": 0.0852, "den": 0.008, "deg": 0.0011, "freq": 0.0036, "hola": 0.0003, "viva": 0.0066, "track": 0.0727}
```